09 Metabolic Vulnerability Prioritization Map

Paper 9 SL companion landing – A public-data prioritization report for glutamine-axis and metabolic vulnerability candidates in lineage-silenced thyroid cancer. This is a research-use map for experimental planning, not a clinical actionability claim.

Status: GO (as map)

Hypothesis-generating only

Not validated SL

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Anchor: Paper 1 DM1 / RAI-lineage axis

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1. 한 줄 결론

Paper 9 is GO only as a metabolic vulnerability prioritization map. 검증된 합성치사 (synthetic lethality) 논문이 아니다. Top wet-lab candidates: **SLC1A5 > GLUD1 > GLS.** lineage-silenced thyroid cancer 의 glutamine-axis 취약성을 우선순위로 매긴 hypothesis-queue 이며, 검증은 wet-lab 에서 시작되어야 한다.

TOP WET-LAB CANDIDATE	STRICT THYROID CRISPR N	GLS BPTES DRUG R	SL CLAIM
SLC1A5	**11**	**−0.015**	**NO**
glutamine transporter, dependency r=0.102 p=5.3e-4 n=1140	underpowered for thyroid-specific inference	p=0.694; weak/non-supportive (drug concordance)	No validated SL; clinical recommendation forbidden

PROBLEM

GLS 단일 SL claim 은 BPTES drug concordance 가 약해서 위험. lineage-silenced thyroid cancer 의 glutamine-axis 전체를 묻는 안전한 frame 이 더 흥미롭다.

IDEA

public dependency / expression / drug screen 3 개 stream 을 합쳐서 candidate 우선순위만 매기는 prioritization map 으로 reframe.

DEFENSE

wet-lab 검증 endpoint 만 명시. SL · 임상 권고 · biomarker claim 은 명시적으로 forbidden 으로 분리.

DECISION

SLC1A5 > GLUD1 > GLS 순서로 wet-lab 1 차 검증 queue. paper 는 "validated SL" 이 아니라 "candidate prioritization".

2. 비전공자 배경

이 페이지는 **합성치사 (synthetic lethality, SL)** 의 후보를 추천하는 지도다. SL 은 두 유전자 중 하나만 고장나면 세포가 사는데, 둘 다 고장나면 세포가 죽는 관계. 암세포가 이미 한쪽이 망가져 있다면, 나머지 한쪽을 약으로 누르면 정상세포는 살고 암세포만 선택적으로 죽는다 – 이것이 SL 기반 항암제의 logic.

갑상선암에서 **lineage-silenced** (= 갑상선다운 정체성을 잃은) state 는 Paper 1 의 DM1 / RAI-lineage axis 로 정의된다. 이 state 가 어떤 metabolic vulnerability 를 노출하는지 – 특히 glutamine 축에서 – 가 Paper 9 의 질문이다.

중요한 framing: **이 페이지는 SLC1A5 / GLUD1 / GLS 가 검증된 SL 표적이라고 말하지 않는다.** public dependency / expression data 가 glutamine-axis 취약성 패턴을 nominate 할 만큼은 강하다는 것이다. 올바른 endpoint 는 prioritization map 이지 treatment recommendation 이 아니다.

2.1 용어 정리

TERM	풀어쓴 설명	이 페이지에서의 역할
Lineage-silenced	갑상선 lineage program (FOXE1/NKX2-1/PAX8/HHEX) 가 collapse 한 state.	biological state variable. Paper 1 의 DM1 axis 와 동일.
Synthetic lethality (SL)	한쪽이 손상 + 다른 쪽 표적 = 선택적 사멸.	이 페이지는 SL 후보만 nominate 하고 검증은 deferred.
Dependency strength	−CRISPR gene effect. 양수 클수록 의존성 강함.	DM1-like state 와 함께 변하는지 본다.
SLC1A5	Glutamine transporter (ASCT2). 세포 안으로 glutamine 을 날라준다.	1 위 wet-lab candidate.
GLUD1	Glutamate dehydrogenase. glutamine → glutamate → α-KG metabolic routing.	2 위. glutaminase blockade 외의 routing 에 주목.
GLS	Glutaminase. glutamine → glutamate.	3 위. dependency / expression 은 있지만 BPTES drug concordance 가 약함.
Artifact risk	signal 이 진짜 metabolic 취약성인지, 단순 proliferation / common essentiality / tissue identity 인지.	모든 candidate 에 risk band 표시.

3. 전체 논리 흐름

FROM PAPER 1

"DM1 = lineage-collapsed state"

FOXE1/NKX2-1/PAX8/HHEX collapse + STAT3/AP-1/DNMT activation. 8-gene RAI compact readout 으로 측정.

→ ask

IN PAPER 9 (THIS MAP)

"Given DM1, what kills it?"

public dependency / expression / drug screen 으로 metabolic vulnerability 후보를 우선순위 매김. 1 차 wet-lab queue 출력.

1 **Start from lineage silencing.** Paper 1 의 differentiation / lineage-silenced axis 를 biological state variable 로 사용. dedifferentiated thyroid-like state 가 metabolic dependency 를 노출하는가가 질문.

2 **Ask dependency question.** public CRISPR screen 에서 lineage silencing 이 강한 cell line 이 glutamine / metabolic gene 에 더 의존적인지 본다.

3 **Remove obvious artifacts.** pan-cancer 와 cancer-type-centered association 을 비교하고, MYC / LDHA / OXPHOS / NAMPT 같은 proliferation / common-essential signal 을 flag.

4 **Convert to wet-lab priority.** dependency / expression / drug metadata / artifact risk / thyroid coverage 5 축으로 ranking. 출력은 experiment queue.

Q1 — 왜 GLS 단일 SL 로 가지 않았나?

BPTES drug concordance 가 r=−0.015 p=0.694 으로 약/non-supportive. 단일 표적 검증 압력 못 견딘다.

Q2 — 왜 SLC1A5 가 1 위?

transporter 라서 upstream 에 있다. cancer-type correction 후에도 dependency association 유지 (r=0.078 p=0.008).

Q3 — 왜 GLUD1 이 2 위?

glutaminase blockade 외에 glutamate routing 으로 분기. 보완적 axis 로 가치.

Q4 — 가장 큰 RISK?

strict thyroid CRISPR n=11 — thyroid-specific inference 통계력 부족. wet-lab 검증 시 lineage-silenced vs lineage-high thyroid 모델 비교 필수.

Q5 — TROP2 / ADC 는 왜 빠졌나?

Paper 9 는 SL/metabolic 만 다룸. TROP2 ADC 는 별도 trajectory (Paper 10 / Phase B drug platform).

FINAL — MAP YES, CLAIM NO.

candidate priority 는 안전. 임상 / 검증된 SL claim 은 명시적으로 forbidden.

4. How To Read This Report

Important framing. This page is not saying that SLC1A5, GLUD1, or GLS are validated synthetic-lethal targets. It says that public dependency/expression data nominate a glutamine-axis vulnerability pattern that is strong enough to justify wet-lab testing. The correct endpoint is a prioritization map, not a treatment recommendation.

Why this analysis was done. GLS alone was not strong enough for a synthetic-lethality claim because drug concordance was weak. The safer and more interesting direction is broader: lineage-silenced thyroid cancer may shift toward glutamine transport and glutamate metabolism vulnerabilities. That is why the page prioritizes **SLC1A5** > **GLUD1** > **GLS**, instead of forcing a GLS-only story.

5. Executive Verdict

PAPER 9 STATUS

GO

As a metabolic vulnerability prioritization map for lineage-silenced thyroid cancer.

BOUNDARY

Not SL-Proof

Not validated synthetic lethality. Not clinical treatment recommendation.

TOP WET-LAB CANDIDATES

SLC1A5 > GLUD1 > GLS

Prioritize transporter/ metabolism axis first; keep GLS as a candidate.

Directionality sanity check: positive dependency_strength means `-CRISPR gene effect`, so a positive association indicates stronger dependency with lineage silencing. For drug matrices, raw AUC is not sensitivity; higher AUC generally means less sensitivity/resistance. Drug summaries therefore use direction-normalized sensitivity as `-AUC`. PRISM LFC had no processed GLS/SLC1A5/GLUD1 metadata match in this pass.

Most important take-home message. Paper 9 is GO only as a metabolic vulnerability prioritization map. It is not GO as a validated synthetic lethality paper. The first wet-lab test should ask whether lineage-silenced thyroid models are selectively more vulnerable to SLC1A5 or GLUD1 perturbation, with GLS kept as a candidate rather than the central proven target.

6. Candidate Ranking Table

This table converts multiple imperfect evidence streams into a practical experimental ranking. A high rank does not mean the target is clinically actionable. It means the candidate survived enough public-data filters to deserve wet-lab validation.

- **Dependency evidence**: whether lineage-silenced models show stronger CRISPR dependency.
- **Expression evidence**: whether the candidate or module tracks the lineage-silenced state at RNA level.
- **Drug response evidence**: whether existing processed drug screens contain a matched compound and a consistent sensitivity direction.
- **Artifact risk**: whether the signal may reflect proliferation, common essentiality, or tissue identity rather than a specific metabolic vulnerability.
- **Thyroid coverage**: whether strict thyroid model overlap is enough for thyroid-specific inference. Here it remains underpowered.

RANK	CANDIDATE	CLASS	DEPENDENCY EVIDENCE	EXPRESSION EVIDENCE	DRUG RESPONSE EVIDENCE	ARTIFACT RISK	THYROI SPECIF COVERA
1	**SLC1A5**	glutamine transport	supportive; r=0.102, p=0.000533, n=1140	weak direction; expression association present	not_testable; no matched processed drug metadata	moderate tissue-context risk	strict depend n=11; underp
2	**GLUD1**	glutamate dehydrogenase	supportive; r=0.0754, p=0.0109, n=1140	weak/low magnitude	not_testable; no matched processed drug metadata	low-to-moderate	strict depend n=11; underp
3	**GLS**	glutaminase	supportive; r=0.0881, p=0.0029, n=1140	supportive; r=0.127, p=1.26e-07, n=1719	BPTES weak/non-supportive; r=-0.0153, p=0.694	moderate tissue-context risk	strict depend n=11; underp
4	MYC glutamine-addiction module	secondary module	supportive module signal	weak	weak/ discouraging drug concordance	proliferation/ common-essential guarded	strict depend n=11; underp
5	SLC7A5	amino-acid transport	watchlist; centered signal only	weak	not_testable	moderate tissue-context risk	strict depend n=11; underp
6	GOT2	transaminase	watchlist; centered signal only	weak	not_testable	low-to-moderate	strict depend n=11; underp
7	LDHA / OXPHOS / NAMPT	comparators	not primary	context/ proliferation-linked	weak or not testable	artifact guarded	strict depend n=11; underp

WHY SLC1A5 IS RANKED FIRST

SLC1A5 is a glutamine transporter, so it sits upstream of glutamine uptake. Its dependency association remains after cancer-type correction, which makes it a

WHAT IS STILL MISSING

The strict thyroid CRISPR overlap is small, and processed drug metadata did not provide a clean SLC1A5/GLUD1 drug test. Therefore the ranking is a stronger first wet-lab candidate than GLS in this public-data pass.

hypothesis queue, not target validation.

7. Key Evidence Cards

Each card separates the strongest public-data signal from the main weakness. The goal is to make the professor-facing message honest: SLC1A5 and GLUD1 are stronger wet-lab priorities, GLS remains biologically plausible, but none of them is proven.

SLC1A5

Meaning: strongest current wet-lab priority because transporter dependency remains visible after pan-cancer correction.

- dependency r=0.102, p=0.000533, n=1140
- pan-cancer corrected dependency r=0.078, p=0.00838
- strict thyroid dependency n=11; underpowered
- artifact risk: moderate tissue-context risk
- verdict: PRIORITIZE_FOR_WETLAB

Needs next: SLC1A5 knockdown/inhibition, glutamine withdrawal, and rescue in lineage-high vs lineage-silenced thyroid models.

GLUD1

Meaning: points to glutamate dehydrogenase/metabolic routing rather than only glutaminase blockade.

- dependency r=0.0754, p=0.0109, n=1140
- pan-cancer corrected dependency r=0.0619, p=0.0365
- strict thyroid dependency n=11; underpowered
- artifact risk: low-to-moderate
- verdict: PRIORITIZE_FOR_WETLAB

Needs next: GLUD1 knockdown or inhibitor testing with viability, proliferation, and rescue controls.

GLS

Meaning: remains plausible, but weak BPTES concordance prevents a GLS-only conclusion.

- dependency r=0.0881, p=0.0029, n=1140
- expression r=0.127, p=1.26e-07, n=1719
- pan-cancer corrected dependency r=0.0808,

WHY GLS WAS DEMOTED

GLS has dependency and expression support, but the available BPTES drug concordance is weak/non-supportive. That means GLS should stay in the experimental set, but the paper should not be framed as "GLS synthetic

p=0.00635
• BPTES drug concordance weak/non-supportive: r=-0.0153, p=0.694
• verdict: KEEP_AS_CANDIDATE, MEDIUM_MINUS
Needs next: test modern GLS inhibitor conditions and glutamine rescue before making any stronger statement.

lethality." The stronger frame is glutamine-axis prioritization.

8. Claim Boundary

The public data support candidate nomination, not clinical translation. This distinction protects the paper from overclaiming and makes the wet-lab plan the natural next step.

STATUS	CLAIMS
ALLOWED	candidate metabolic vulnerability · research-use prioritization · wet-lab validation candidates
WEAK	thyroid-specific inference, because strict thyroid CRISPR overlap is small (n=11)
FORBIDDEN	validated synthetic lethality · clinical treatment recommendation · patient selection biomarker

9. Wet-Lab Validation Plan

Critical experimental question. Do lineage-silenced thyroid models show greater functional vulnerability to glutamine-axis perturbation than lineage-high thyroid models? This is the key experiment needed before any stronger mechanistic or therapeutic language is justified.

Experimental next gate. Compare lineage-high vs lineage-silenced thyroid models. Test SLC1A5 inhibition or knockdown, GLUD1 knockdown or inhibition, GLS inhibitor plus glutamine withdrawal, glutamine rescue assay, MYC/proliferation artifact control, and optional isotope tracing.

MINIMUM VALIDATION PACKAGE

- Pick at least two lineage-silenced and two lineage-high thyroid models.
- Measure baseline SLC1A5, GLUD1, GLS expression and glutamine-dependence.
- Run knockdown or inhibitor assays with matched viability/proliferation readouts.
- Add glutamine rescue or media modulation to prove metabolic specificity.

ARTIFACT CONTROLS

- Control for MYC/proliferation so the signal is not just faster-growing cells dying.
- Compare LDHA, OXPHOS, and NAMPT as comparator axes, not primary claims.
- Do not interpret thyroid-only public estimates as definitive because strict overlap is n=11.
- Optional isotope tracing can test whether glutamine carbon/nitrogen use actually changes.

10. Professor Summary in Korean

> 교수님, Paper 9 Sprint 2 결과를 보면 GLS 단일 synthetic lethality 논문으로 가는 것은 위험하지만, lineage-silenced thyroid cancer의 glutamine-axis metabolic vulnerability를 우선순위화하는 논문 방향은 가능성이 있습니다.
>
> GLS는 dependency와 expression association은 관찰되지만 BPTES drug response concordance가 약해서 단독 validated target으로 주장하기 어렵습니다. 반면 SLC1A5와 GLUD1은 cancer-type correction 후에도 dependency association이 유지되고, 특히 SLC1A5는 glutamine transporter로서 lineage-silenced state와 가장 안정적으로 연결되어 wet-lab 1차 검증 후보로 우선순위가 높습니다.
>
> 따라서 Paper 9의 안전한 claim은 "validated synthetic lethality"가 아니라 "lineage-silenced thyroid cancer에서 glutamine-axis vulnerability candidates를 prioritization했다"입니다. 후속 검증은 SLC1A5, GLUD1, GLS를 중심으로 glutamine withdrawal, inhibitor sensitivity, rescue assay를 설계하는 것이 적절합니다.

> **Paper 9 is GO** as a metabolic vulnerability prioritization map. It is **not GO** as validated synthetic lethality. Top wet-lab candidates: **SLC1A5** > **GLUD1** > **GLS**. Paper 9는 검증된 합성치사 논문이 아니라, lineage-silenced thyroid cancer의 대사 취약성 후보 우선순위 지도입니다. 1차 wet-lab 후보는 SLC1A5 > GLUD1 > GLS입니다.

11. Claim boundary table

CLAIM	STATUS
Glutamine-axis candidate prioritization map	YES
SLC1A5 / GLUD1 / GLS as wet-lab queue (research-use)	YES
Thyroid-specific inference (strict CRISPR n=11)	WEAK
Validated synthetic lethality	NO
Clinical treatment recommendation	NO
Patient selection biomarker	NO

Next action: Paper 1 in print → marathon close → wet-lab Tier 1 (SLC1A5 / GLUD1 inhibition + glutamine rescue in lineage-silenced vs lineage-high thyroid models).

12. 데이터 / 다운로드 · 자매 페이지

Paper 9 SL companion landing → Paper 9 plan → Paper 9 first-pass → Paper 10 atlas → Paper 11 pan-cancer transfer → Paper 12 network. Paper 11/12 는 Paper 10 의 36-candidate panel 을 downstream 으로 사용한다.

📋 PAPER 9 PLAN
Strategic / Ruppin-style design doc (post-marathon)

📊 PAPER 9 FIRST-PASS
Local-data-only triangulation results

📊 PAPER 10 ATLAS
17 figures + 9 sortable tables (visualization companion)

📋 PAPER 11 PANCANCER
DM1 transfer atlas across 33 TCGA lineages

📜 PAPER 12 NETWORK
Candidate × candidate co-expression network

📁 PHASE B PLATFORM
TROP2 ADC carve-out (non-SL)

← PAPER 1 ANCHOR
DM1 / RAI-lineage axis 정의

📄 PAPER 9 PORTFOLIO CARD
One-page detail card

← HUB INDEX
8-papers portfolio

Status. Paper 9 = GO as metabolic vulnerability prioritization map. Not GO as validated SL. Wet-lab queue: SLC1A5 > GLUD1 > GLS. Anchor: Paper 1 DM1 axis. Sister pages: Paper 10/11/12 (visualization, pan-cancer transfer, network).

Live URL. `http://40.82.129.113:8012/papers_hub_2026_05_04/paper9.html`